COMBINATORIAL ANALYSIS

Lect. İsmail BAŞOĞLU, Ph.D.

Özyeğin University

February 2, 2024

1. Preliminary Definitions
   - What is Combinatorial Analysis?
   - Countable Sets
   - Experiment and Outcomes of An Experiment

1. Preliminary Definitions
   - What is Combinatorial Analysis?
   - Countable Sets
   - Experiment and Outcomes of An Experiment
2. Principles of Counting
   - Basic Principle of Counting
   - Generalized Principle of Counting

1. Preliminary Definitions
   - What is Combinatorial Analysis?
   - Countable Sets
   - Experiment and Outcomes of An Experiment
2. Principles of Counting
   - Basic Principle of Counting
   - Generalized Principle of Counting
3. Permutations
   - What Is A Permutation?
   - Permutations of Distinguishible Objects
   - Permutations With Indistinguishible Objects

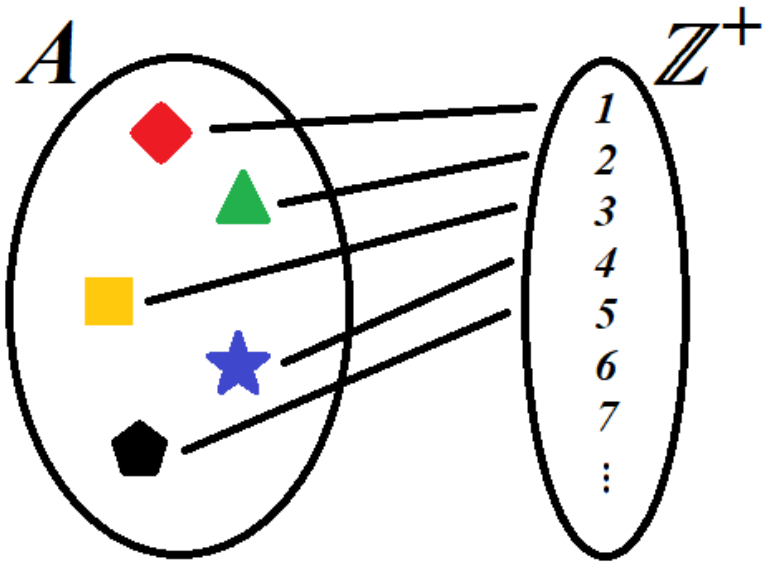
1. Preliminary Definitions
   - What is Combinatorial Analysis?
   - Countable Sets
   - Experiment and Outcomes of An Experiment
2. Principles of Counting
   - Basic Principle of Counting
   - Generalized Principle of Counting
3. Permutations
   - What Is A Permutation?
   - Permutations of Distinguishible Objects
   - Permutations With Indistinguishible Objects
4. Combinations
   - What Is A Combination?
   - The Binomial Theorem
   - Multinomial Coefficients

What is Combinatorial Analysis?

- Before talking about probability theory and its applications, we need to start from the simple!
  - We need to learn how to count!
- **Combinatorial Analysis:** The mathematical theory of counting.

Countable Sets

- When we count something, we always count the number of elements in a set.
- A set is **countable** if its elements can be put in a one-to-one correspondence with the sequence of positive integers, $\mathbb{Z}^+$.



Countable vs. Uncountable Sets

- Examples to countable sets:
  - The earthquakes happening in 2018 are countable.
  - The marbles in an urn are countable.
  - The human population on earth is countable.
- Examples to uncountable sets:
  - The water in a dam is a set with uncountable number of elements.
  - The flour used in a loaf of bread is a set with uncountable number of elements.
  - The set of real numbers: $\mathbb{R}$
  - Unit interval: $\{x : x \in \mathbb{R}, 0 \leq x \leq 1\}$

Experiment and Outcome

- **Experiment:** A natural phenomenon whose occurrence can be observed or a procedure that can be performed for examination purposes.
- **Outcome:** The consequence of an experiment.

Random Experiments

- **Random Experiment:** An experiment that exhibits an inherent variation in outcomes, i.e. the outcome of which is not predictable in advance.
  - Throwing an egg from a building is an experiment with a predictable outcome.
  - Drawing two cards from a 52-deck is a **random experiment** with many possible outcomes.
  - How many is a question that is dealt by **combinatorial analysis**.
- Many problems in probability theory can be solved simply by counting the number of different ways that a certain experiment can result.

Simple Probability Example

Q1. A communication system is to consist of four seemingly identical antennas that are to be lined up in a linear order. The resulting system will then be able to receive all incoming signals - and will be called functional - as long as no two consecutive antennas are defective. It turns out that exactly two of the four antennas are defective.

a. With respect to operating status of the antennas, how many different orderings are possible?
b. How many of the above orderings will yield a functional system?
c. What is the probability that the resulting system will be functional?

Simple Probability Example (cont'd)

A1. Let "W" and "D" respectively represent a working and a defective antenna.

a. The following list contains all possible orderings of these four antennas, two of which are defective.

$$\begin{array}{lcccc} 1: & W & W & D & D \\ 2: & W & D & W & D \\ 3: & W & D & D & W \\ 4: & D & W & W & D \\ 5: & D & W & D & W \\ 6: & D & D & W & W \end{array}$$

Simple Probability Example (cont'd)

b. Among the above orderings, 1st, 3rd and 6th orderings are non-functional, as in these orderings defective antennas are consecutive.

$$\begin{array}{lcccc} 1: & W & W & \boldsymbol{D} & \boldsymbol{D} \\ 2: & W & D & W & D \\ 3: & W & \boldsymbol{D} & \boldsymbol{D} & W \\ 4: & D & W & W & D \\ 5: & D & W & D & W \\ 6: & \boldsymbol{D} & \boldsymbol{D} & W & W \end{array}$$

c. Out of six possible orderings (systems), three are functional. Thus, the probability of having a functional system is $\frac{3}{6} = 0.5$.

Basic Principle of Counting

- Suppose that two experiments are to be performed.
  - Experiment 1 can result in any one of m possible outcomes.
  - For each outcome of experiment 1, there are n possible outcomes of experiment 2.
- Then, together, there are $m \times n$ possible outcomes of the two experiments.

Basic Principle of Counting: Example

Q2. A small community consists of 10 women, each of whom has 3 children. If one woman and one of her children are to be chosen as mother and child of the year, how many different choices are possible?

Basic Principle of Counting: Example

Q2. A small community consists of 10 women, each of whom has 3 children. If one woman and one of her children are to be chosen as mother and child of the year, how many different choices are possible?

A2. By regarding the choice of the woman as the outcome of the first experiment and the subsequent choice of one of her children as the outcome of the second experiment, we see from the basic principle that there are $10 \times 3 = 30$ possible choices.

Generalized Principle of Counting

- When there are more than two experiments to be performed, the basic principle of counting can be generalized.
- Suppose now, r experiments that are to be performed, such that:
  - the first experiment may result in any of n_1 possible outcomes,
  - for each of these n_1 possible outcomes, there are n_2 possible outcomes of the second experiment,
  - for each of the possible outcomes of the first two experiments, there are n_3 possible outcomes of the third experiment
  - and the remaining experiments continue in this manner.
- Then, altogether, there is a total of

$$n_1 \times n_2 \times \ldots \times n_r$$

possible outcomes of the r experiments.

Generalized Principle of Counting: Example

Q3. A college planning committee consists of 3 first-year, 4 second-year, 5 third-year, and 2 fourth-year students. A subcommittee of 4, consisting of 1 person from each class, is to be chosen. How many different subcommittees are possible?

Generalized Principle of Counting: Example

Q3. A college planning committee consists of 3 first-year, 4 second-year, 5 third-year, and 2 fourth-year students. A subcommittee of 4, consisting of 1 person from each class, is to be chosen. How many different subcommittees are possible?

A3. We may regard the choice of a subcommittee as the combined outcome of the four separate experiments of choosing a single representative from each of the classes. It then follows from the generalized principle that there are

$$3 \times 4 \times 5 \times 2 = 120$$

possible subcommittees.

Generalized Principle of Counting: Example II

Q4. Suppose we are flipping a coin n times, consecutively. How many different outcomes are possible if each coin flip results either with heads (H) or tails (T)?

Generalized Principle of Counting: Example II

Q4. Suppose we are flipping a coin n times, consecutively. How many different outcomes are possible if each coin flip results either with heads (H) or tails (T)?

A4. Simply, each coin flip is considered as an experiment with two outcomes. According to the generalized principle of counting, the number of all outcomes is calculated as:

$$\underbrace{2 \times 2 \times \ldots \times 2}_{n\,\text{times}} = 2^n$$

Generalized Principle of Counting: Example III

Q5. We design a license plate that consists of seven characters, first three being capital letters in Turkish alphabet excluding "Ğ", the last four being figures excluding zero.

a. How many license plates would be possible?
b. How many license plates would be possible if repetition among letters were prohibited?
c. How many license plates would be possible if repetition among letters and figures were prohibited?

Generalized Principle of Counting: Example III (cont'd)

A5. There are 28 letters in Turkish alphabet excluding "Ğ" and there are nine figures excluding zero.

Generalized Principle of Counting: Example III (cont'd)

A5. There are 28 letters in Turkish alphabet excluding "Ğ" and there are nine figures excluding zero.

a. $28 \times 28 \times 28 \times 9 \times 9 \times 9 \times 9 = 144,027,072$

Generalized Principle of Counting: Example III (cont'd)

A5. There are 28 letters in Turkish alphabet excluding "Ğ" and there are nine figures excluding zero.

a. $28 \times 28 \times 28 \times 9 \times 9 \times 9 \times 9 = 144,027,072$

b. When repetition among letters is prohibited:

$$28 \times 27 \times 26 \times 9 \times 9 \times 9 \times 9 = 128,963,016$$

Generalized Principle of Counting: Example III (cont'd)

A5. There are 28 letters in Turkish alphabet excluding "Ğ" and there are nine figures excluding zero.

a. $28 \times 28 \times 28 \times 9 \times 9 \times 9 \times 9 = 144,027,072$

b. When repetition among letters is prohibited:

$$28 \times 27 \times 26 \times 9 \times 9 \times 9 \times 9 = 128,963,016$$

c. When repetition of any character is prohibited:

$$28 \times 27 \times 26 \times 9 \times 8 \times 7 \times 6 = 59,439,744$$

What Is A Permutation?

- A permutation is a rearrangement of the objects in a set into some sequence or order.
  - The objects: $\{\blacksquare, \triangledown, \circ, \blacktriangle\}$.
  - A permutation: $\circ, \triangledown, \blacktriangle, \blacksquare$
  - Another permutation (ordering): $\triangledown, \blacksquare, \blacktriangle, \circ$

Permutations of Distinguishible Objects

- If we have n different objects, how many permutations (possible orderings) do we have?
  - A permutation starts with n possible candidates.
  - Once we position one of these objects, we will have $n-1$ remaining candidates for the second position in this permutation.
  - Once the second object is positioned, we will have $n-2$ candidates for the third position.
  - $\ldots$
- By the generalized principle of counting, there are

$$n \times (n-1) \times (n-2) \times \ldots \times 3 \times 2 \times 1 = n!$$

different permutations of n objects.

Permutations of Distinguishible Objects: Example

Q6. A statistics class consists of 6 men and 4 women. An examination is given, and the students are ranked according to their performance. Assume that no two students obtain the same score.

a. How many different rankings are possible?
b. If the men are ranked just among themselves and the women just among themselves, how many different rankings are possible?

Permutations of Distinguishible Objects: Example (cont'd)

A6. Each ranking corresponds to a particular ordered arrangement of the 10 people.

a. The answer to first part is:

$$10! = 3,628,800$$

b. Since there are $6!$ possible rankings of the men among themselves and $4!$ possible rankings of the women among themselves, it follows from the basic principle that there are

$$6! \times 4! = 720 \times 24 = 17280$$

possible rankings in this case.

Permutations of Distinguishible Objects: Example II

Q7. Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

Permutations of Distinguishible Objects: Example II

Q7. Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

A7. There are $4! = 24$ possible orderings of the subjects on the shelf and for each subject ordering, there are $4!3!2!1! = 288$ arrangements within books. Hence, there are

$$4!4!3!2!1! = 24 \times 288 = 6912$$

possible orderings.

Permutations With Indistinguishible Objects

- Suppose now, there are n objects in the set. However, some of these objects are indistinguishable from each other.
- Assume further, there are at most r of these objects that are different from each other:
  - There are n_1 objects of the first type,
  - There are n_2 objects of the second type,
  - $\ldots$
  - There are n_r objects of the rth type.

$$n_1 + n_2 + \ldots + n_r = n$$

- In that case, there are

$$\frac{n!}{n_1!n_2!\ldots n_r!}$$

different permutations of these objects.

Permutations With Indistinguishible Objects: Example

Q8. A chess tournament has 10 competitors, of which 4 are Russian, 3 are from the United States, 2 are from Great Britain, and 1 is from Brazil. If the tournament result lists just the nationalities of the players in the order in which they placed, how many outcomes are possible?

Permutations With Indistinguishible Objects: Example

Q8. A chess tournament has 10 competitors, of which 4 are Russian, 3 are from the United States, 2 are from Great Britain, and 1 is from Brazil. If the tournament result lists just the nationalities of the players in the order in which they placed, how many outcomes are possible?

A8. There are

$$\frac{10!}{4!3!2!1!} = \frac{10\times9\times8\times7\times6\times5\times4!}{4!3!2!1!} = 10\times9\times4\times7\times5 = 12600$$

possible outcomes.

Permutations With Indistinguishible Objects: Example

Q9. How many different signals, each consisting of 11 flags hung in a line, can be made from a set of 5 white flags, 4 red flags, and 2 blue flags if all flags of the same color are identical?

Permutations With Indistinguishible Objects: Example

Q9. How many different signals, each consisting of 11 flags hung in a line, can be made from a set of 5 white flags, 4 red flags, and 2 blue flags if all flags of the same color are identical?

A9. There are

$$\frac{11!}{5!4!2!} = \frac{11\times10\times9\times8\times7\times6\times5!}{5!4!2!} = 11\times10\times9\times7 = 6930$$

different signals.

What Is A Combination?

- A combination is a way of selecting objects from a collection, such that, unlike permutations, the order of selection does not matter.
  - Permutation = A possible ordering
  - Combination = A possible group selection
- Consider the set $\{\blacksquare, \triangledown, \circ, \blacktriangle\}$.
  - $\{\circ, \blacktriangle, \triangledown\}$ is a 3-element combination from the above set.
  - $\{\blacktriangle, \triangledown, \circ\}$ is the same 3-element combination (not another combination).
  - $\{\blacktriangle, \circ, \blacksquare\}$ is another 3-element combination from the above set.

Combinations: Example

Q10. How many different groups of 3 could be selected from the 5 items A, B, C, D, and E?

Combinations: Example

Q10. How many different groups of 3 could be selected from the 5 items A, B, C, D, and E?

A10. There are 5 ways to select the initial item, 4 ways to then select the next item, and 3 ways to select the final item, there are:

$$5 \times 4 \times 3 = 60$$

ways of selecting the group of 3 when the order in which the items are selected is relevant.

Since every group of 3 items can be selected in $3! = 6$ different permutations, it follows that the total number of groups that can be formed is:

$$\frac{5 \times 4 \times 3}{3!} = 10$$

Combinations: Example (cont'd)

- When the order of selection is relevant:

ABC	ABD	ABE	ACD	ACE	ADE	BCD	BCE	BDE	CDE
ACB	ADB	AEB	ADC	AEC	AED	BDC	BEC	BED	CED
BAC	BAD	BAE	CAD	CAE	DAE	CBD	CBE	DBE	DCE
BCA	BDA	BEA	CDA	CEA	DEA	CDB	CEB	DEB	DEC
CAB	DAB	EAB	DAC	EAC	EAD	DBC	EBC	EBD	EDC
CBA	DBA	EBA	DCA	ECA	EDA	DCB	ECB	EDB	ECD

- When the order of selection is irrelevant (10 combinations):

ABC	ABD	ABE	ACD	ACE
ADE	BCD	BCE	BDE	CDE

Number of Combinations

- We are often interested in determining the number of different groups of k objects that could be formed from a total of n objects.
  - The number of different ways that a group of k objects could be selected from n objects when the order of selection is relevant is:
$$n \times (n-1) \times \ldots \times (n-k+1)$$
  - In this case, however, each group of k objects will be counted $k!$ times in this count.
- Therefore, **the number of different groups** of k objects that could be formed from a total of n objects is:
$$\frac{n(n-1)\ldots(n-k+1)}{k!} = \frac{n(n-1)\ldots(n-k+1)}{k!}\frac{(n-k)!}{(n-k)!} = \frac{n!}{k!(n-k)!}$$
- **Number of k-combinations:** Number of k-element subsets in a set of n elements.

Combination Terminology

- For non-negative integers k and n such that $k \leq n$, we define the combination terminology as

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

  which represents the number of different groups of size k that could be selected from a set of n objects when the order of selection is not considered relevant[1].
- The terminology is often read aloud as "n **choose** k".

[1] When $n = k$, do not forget that $0! = 1$.

Combinations: Example II

Q11. A committee of 3 is to be formed from a group of 20 people. How many different committees are possible?

Combinations: Example II

Q11. A committee of 3 is to be formed from a group of 20 people. How many different committees are possible?

A11. There are

$$\binom{20}{3} = \frac{20!}{3!\,(20-3)!} = \frac{20!}{3!17!} = \frac{20 \times 19 \times 18}{6} = 1140$$

possible committees.

Combinations: Example III

Q12. From a group of 5 women and 7 men:

a. How many different committees consisting of 2 women and 3 men can be formed?

b. Suppose 2 of the men are feuding and refuse to serve on the committee together. Answer the same question.

Combinations: Example III (cont'd)

A12. Answers are as follows:

a. There are:

- $\binom{5}{2}$ possible groups of 2 women and
- $\binom{7}{3}$ possible groups of 3 men.

It follows from the basic principle of counting that there are:

$$\binom{5}{2}\binom{7}{3} = \frac{5!}{2!3!}\frac{7!}{3!4!} = \left(\frac{5\times 4}{2}\right)\left(\frac{7\times 6\times 5}{6}\right) = 350$$

possible committees.

Combinations: Example III (cont'd)

b. Let us count the number of committees in which these two men exist together. If these two men already exist, we need to:

- choose one more man out of the remaining five men and
- two women out of the five women.

That makes:

$$\binom{5}{2}\binom{5}{1} = \frac{5!}{2!3!}\frac{5!}{1!4!} = 50$$

committees in which these feuding men exist together.

The question asks the number of committees in which the feuding men does not exist together, we say $350 - 50 = 300$ committees.

The Binomial Theorem

- **The Binomial Theorem:** Let x and y be arbitrary real numbers ($x, y \in \mathbb{R}$) and n be a non-negative integer. Then:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

- **Proof:** The proof is by induction, but will not be shown here. The outline of the proof is as follows.
  - When $n = 1$, the equation holds as $x + y = x + y$.
  - We assume the equation holds for non-negative integer n.
  - Under this assumption, we show that the equation also holds for $n + 1$. The proof, then, will be complete.

The Binomial Theorem: Example

Q13. Expand:

a. $(x+y)^3$ and
b. $(x+y)^4$

using the binomial theorem.

The Binomial Theorem: Example (cont'd)

A13. Answers are as follows:

a. The expansion of $(x+y)^3$ is as follows:

$$\begin{aligned}(x+y)^3 &= \sum_{k=0}^{3} \binom{3}{k} x^k y^{3-k} \\ &= \binom{3}{0} x^0 y^{3-0} + \binom{3}{1} x^1 y^{3-1} + \binom{3}{2} x^2 y^{3-2} + \binom{3}{3} x^3 y^{3-3} \\ &= y^3 + 3xy^2 + 3x^2y + x^3\end{aligned}$$

b. The expansion of $(x+y)^4$ is as follows:

$$\begin{aligned}(x+y)^4 &= \sum_{k=0}^{4} \binom{4}{k} x^k y^{4-k} \\ &= \binom{4}{0} x^0 y^{4-0} + \binom{4}{1} x^1 y^{4-1} + \binom{4}{2} x^2 y^{4-2} + \binom{4}{3} x^3 y^{4-3} + \binom{4}{4} x^4 y^{4-4} \\ &= y^4 + 4xy^3 + 6x^2y^2 + 4x^3y + x^4\end{aligned}$$

The Binomial Theorem: Example II

Q14. How many subsets are there of a set consisting of n elements?

The Binomial Theorem: Example II

Q14. How many subsets are there of a set consisting of n elements?

A14. We know that for a set of n elements, the number of k-element subsets are $\binom{n}{k}$.

Then, the number of all subsets is:

$$\sum_{k=0}^{n} \binom{n}{k} = \sum_{k=0}^{n} \binom{n}{k} (1)^k (1)^{n-k} = (1+1)^n = 2^n$$

which can be justified by the binomial theorem.

Multinomial Coefficients

- Suppose a set of n different objects is to be divided into r distinct groups of respective sizes $n_1, n_2, \ldots, n_r$, where:

$$n_1 + n_2 + \ldots + n_r = n$$

- How many different divisions (groupings) are possible?

Multinomial Coefficients (cont'd)

- Let us examine the selection procedure.
  - There are $\binom{n}{n_1}$ different ways of selecting the first group.
  - For each selection of the first group, there are $\binom{n-n_1}{n_2}$ different ways of selecting the second group.
  - For each selection of the first and second group, there are $\binom{n-n_1-n_2}{n_3}$ different ways of selecting the third group.
  - ...
  - The last group will contain the remaining n_r objects. There will be a single option that is $\binom{n_r}{n_r}$.

Multinomial Coefficients (cont'd)

- We will use the generalized principle of counting. In total, there are

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n_r}{n_r}$$

different selection options.

- Expanding combination formulas, we get:

$$\frac{n!}{n_1!(n-n_1)!}\frac{(n-n_1)!}{n_2!(n-n_1-n_2)!}\frac{(n-n_1-n_2)!}{n_3!(n-n_1-n_2-n_3)!}\cdots\frac{n_r!}{n_r!0!}=\frac{n!}{n_1!n_2!n_3!\ldots n_r!}$$

Multinomial Coefficient Terminology

- For non-negative integers $n_1, n_2, \ldots, n_r$ and

$$n = n_1 + n_2 + \ldots + n_r,$$

we define the multinomial coefficient terminology

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1! n_2! n_3! \ldots n_r!}$$

which represents the number of different possibilities when we need to select more than one group of objects out of many objects[2].

[2] It is no coincidence that the formula in page 23 and this are the same. See page 21 of the textbook of Ross (2020).

Multinomial Coefficients: Example

Q15. A police department in a small city consists of 10 officers. If the department policy is to have 5 of the officers patrolling the streets, 2 of the officers working full time at the station, and 3 of the officers on reserve at the station, how many different divisions of the 10 officers into the 3 groups are possible?

Multinomial Coefficients: Example

Q15. A police department in a small city consists of 10 officers. If the department policy is to have 5 of the officers patrolling the streets, 2 of the officers working full time at the station, and 3 of the officers on reserve at the station, how many different divisions of the 10 officers into the 3 groups are possible?

A15. There are

$$\binom{10}{5,3,2} = \frac{10!}{5!3!2!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5!}{5! \times 3 \times 2 \times 1 \times 2 \times 1} = 2520$$

possible divisions.

Multinomial Coefficients: Example II

Q16. A classroom of 20 students will be divided into four equal-sized project groups. How many different divisions are possible?

Multinomial Coefficients: Example II

Q16. A classroom of 20 students will be divided into four equal-sized project groups. How many different divisions are possible?

A16. There are

$$\binom{20}{5,5,5,5} = \frac{20!}{5!5!5!5!} = 11,732,745,024$$

possible divisions. Note that, the above calculation can be very hard unless a calculator or a computer software is used.

END OF LECTURE SLIDE 01